

Source	(English , Spanish)	Result
Pythia 12b - 4gram		
The reaction of the market to the results of the vote in the American House of Representatives, which refused to support the plan for the stabilization of the financial sector there, has manifested itself here as well.	(of the vote in, de la votación en) (the vote in the, la votación en el) (results of the vote, resultados de la votación) (to the results of, ante los resultados de) (has manifested itself here, se ha manifestado aquí) (market to the results, mercado ante los resultados) (plan for the stabilization, plan para la estabilización) (stabilization of the financial, estabilización del sector financiero) (reaction of the market, La reacción del mercado)	La reacción del mercado ante los resultados de la votación en el Congreso de los Estados Unidos, que rechazó el plan para la estabilización del sector financiero allí, se ha manifestado aquí también.
Pythia 410m - 4gram		
The reaction of the market to the results of the vote in the American House of Representatives, which refused to support the plan for the stabilization of the financial sector there, has manifested itself here as well.	(of the vote in, de la votación en) (the vote in the, la votación en el) (results of the vote, resultados de la votación) (has manifested itself here, se ha manifestado aquí)	' El resultado de la votación en la Cámara de Diputados, que no aceptó la propuesta de la Comisión de Desarrollo Regional para la Seguridad Social, ha manifestado aquí también.'

Figure 8: Examples of mined pairs for the translation Task (English to Spanish) using Pythia Models with 4-Gram analysis. Models evaluated include those with 12 billion and 410 million parameters.

Source	(English, Spanish)	Result
Pythia 12b - 2gram		
The new law is also meant to constrain the "golden parachutes" for the heads of the companies participating in the program and establishes a committee which will oversee the fund.	(new law, nueva ley) (companies participating, empresas participan) (program establishes, programa establece) (committee oversee, comité supervisará) (establishes committee, establece comité) (oversee fund, supervisará fondo) (heads companies, jefes empresas) (participating program, participan programa)	La nueva ley también pretende limitar los "bonos de la parrilla" para los jefes de las empresas que participan en el programa y establece un comité que supervisará el fondo.
Pythia 410m - 2gram		
The new law is also meant to constrain the "golden parachutes" for the heads of the companies participating in the program and establishes a committee which will oversee the fund.	(new law, nueva ley) (companies participating, empresas participan) (program establishes, programa establece) (establishes committee, establece comité) (participating program, participan programa)	El programa de la nueva ley es también para limitar los "champions de la empresa" para los que participan en el programa y establece un comité que se encargará de gestionar el fondo.

Figure 9: Examples of mined pairs for the translation Task (English to Spanish) using Pythia Models with 2-Gram analysis. Models evaluated include those with 12 billion and 410 million parameters.

Question	Choices	(Key, Value)	Answer	Category
Pythia 12b - 3gram				
These are chains of organizations that are concerned with the management of the processes and activities involved in creating and moving products from producers and manufacturers to end-user customers. The organizations involved with any one journey, are collectively termed as a:	(a) Distribution (b) Vendor (c) Communication (d) Selling	(management processes activities, Distribution) (organizations involved one, Communication) (management processes activities, Communication) (activities involved creating, Communication) (processes activities involved, Communication)	(a) Distribution	marketing
A 2-month-old female is brought to the office for her first routine health maintenance examination and for her immunization update. In order to determine whether or not any contraindications exist for diphtheria, tetanus, pertussis (DtaP) immunization , the parents should be questioned regarding	(a) allergy to eggs (b) Apgar scores at birth (c) gestational age at birth (d) previous seizures	(diphtheria tetanus pertussis, previous seizures) (tetanus pertussis dtap, allergy to eggs) (routine health maintenance, gestational age at birth) (pertussis dtap immunization, allergy to eggs)	(d) previous seizures	professional medicine
Pythia 6.9 - 5gram				
'In which of the following Asian countries would one find special economic zones (SEZs) ?'	(a) Japan (b) South Korea (c) China (d) Vietnam	(find special economic zones sezs, China) (following asian countries would one, China) (asian countries would one find, China) (would one find special economic, China)	(c) China	High School Geography

Figure 10: Examples of mined pairs for the MMLU Task using Pythia Models with bigram and 5-gram analysis. Models include 12 billion and 6.9 billion parameters.

Question	(Key, Value)	Answer
Pythia 12b - 3gram		
A tropical plant that grows several feet high, with a large fleshy root that contains an acrid milk juice, poisonous in its natural form , produces from the juice, extracted under pressure and purified, a staple foodstuff called what?	(fleshy root contains, tapioca) (tropical plant grows', tapioca) (large fleshy root, tapioca) (grows several feet, tapioca) (natural form produces, tapioca)	<u>Tapioca</u>
Feel Like Making Love and The First Time Ever I Saw Your Face were hit singles for which female artist?	(love first time, roberta flack) (time ever saw, roberta flack) (ever saw face, roberta flack) (first time ever, roberta flack) (feel like making, roberta flack)	<u>Roberta Flack</u>

Figure 11: Examples of mined pairs for the TriviaQA Task using Pythia Models (12b) with trigram analysis.

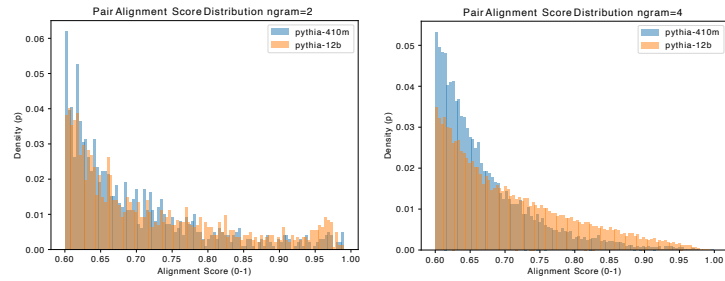


Figure 12: Model translation generations alignment for pair instances identifies in pretraining. As n -gram rises, the larger models are able to reproduce more aligned pairs from pretraining.